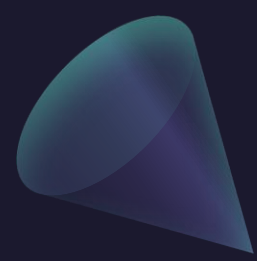


“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 26

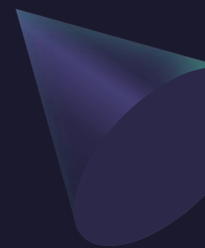


Any Reminders? Any Questions?

- Teaching evaluations are open!
- Exam 3: Friday May 1 – Ch 13.1-13.3 and 16.1-16.8
- Office hours Mondays 1-2pm online
- Review for Exam 3 during class on Thursday!
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104



ALVARO: Start the recording!



“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Stokes' Theorem

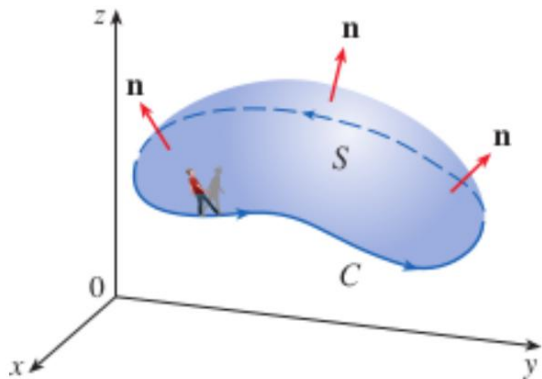


Stokes' Theorem

Stokes' Theorem

Let S be an oriented piecewise-smooth surface that is bounded by a simple, closed, piecewise-smooth boundary curve C with positive orientation. Let $\mathbf{F}$ be a vector field whose components have continuous partial derivatives on an open region in $\mathbb{R}^3$ that contains S . Then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$$



$$\begin{aligned} \nabla \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} \\ &= \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k} \\ &= \text{curl } \mathbf{F} \end{aligned}$$

Sir
George Stokes
FRS



Stokes, 1860s

35th **President of the Royal Society**

In office
1885–1890

Preceded by [Thomas Huxley](#)

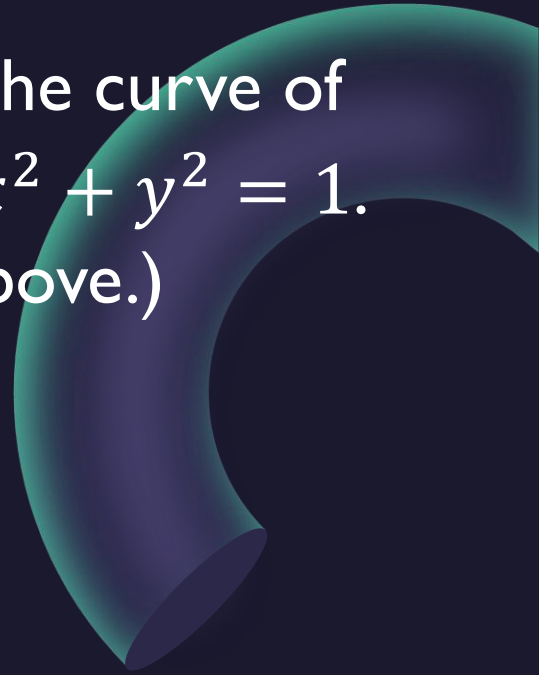
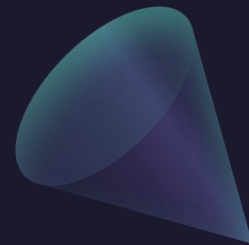
Succeeded by [Lord Kelvin](#)

Personal details

Born George Gabriel Stokes
13 August 1819
[Skreen](#), Ireland, United Kingdom of Great Britain and Ireland

Died 1 February 1903 (aged 83)
[Cambridge](#), England, United Kingdom of Great Britain and Ireland

Example: Evaluate $\int_C F \cdot dr$ where $F = (-y^2, x, z^2)$ and C is the curve of intersection of the plane $y + z = 2$ and the cylinder $x^2 + y^2 = 1$. (C is oriented counterclockwise when viewed from above.)

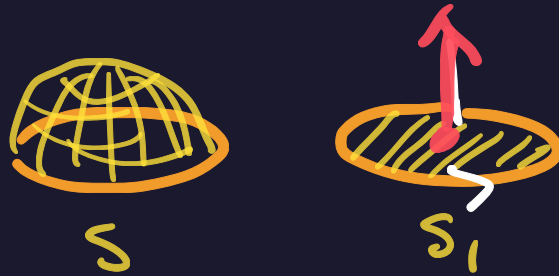
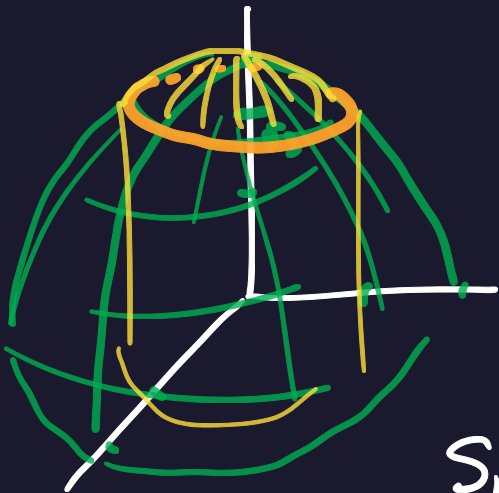


Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (xz, yz, xy)$ and S is the part of the sphere of radius 2 centered at the origin that is inside the cylinder $x^2 + y^2 = 1$ and above the xy -plane.

Alternative solution: Change S to a disk!

Stokes!

$$\iint_S \text{curl}(F) \, dS = \iint_{S_1} \text{curl}(F) \, dS$$



Parametrize S_1 instead!

$$S_1: x^2 + y^2 \leq 1, z = \sqrt{3}$$

$$r(u, v) = (u, v, \sqrt{3}) \quad r_u, r_v, r_u \times r_v, \quad n = (0, 0, 1)$$

$$D = \{(x, y) : x^2 + y^2 \leq 1\}$$

Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (xz, yz, xy)$ and S is the part of the sphere of radius 2 centered at the origin that is inside the cylinder $x^2 + y^2 = 1$ and above the xy -plane.

Alternative solution: Change S to a disk! $r(u,v) = (u, v, \sqrt{3})$

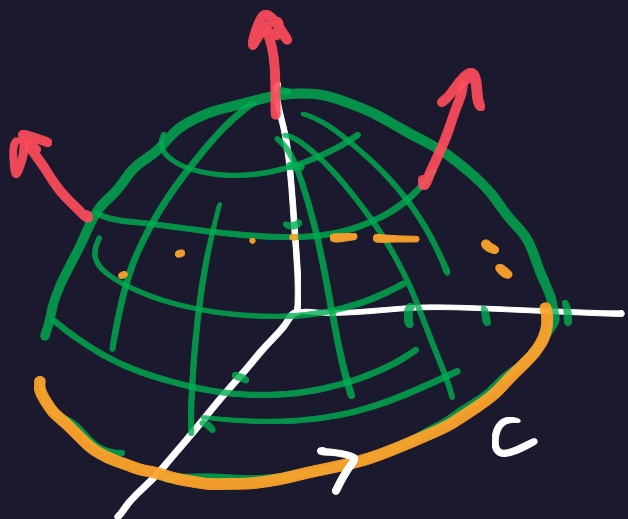
$$\text{Curl}(F) = \nabla \times F = \dots = (x-y, x-y, 0)$$

$$\nabla \times F(r(u,v)) = (u-v, u-v, 0)$$

$$\iint_S \text{curl} F \cdot dS = \iint_D \nabla \times F(r(u,v)) \cdot (0, 0, 1) dA = \iint_D 0 \cdot dA = \boxed{0}$$

$$\nabla \times F(r(u,v)) = (\sim, -, 0) \cdot (0, 0, 1) = 0$$

Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (2y \cos(z), e^x \sin(z), xe^y)$ and S is the semisphere $x^2 + y^2 + z^2 = 9, z \geq 0$ and oriented upwards.



$$\iint_S \text{curl}(F) \, dS = \oint_C F \cdot dr$$

STOKES

$$C = \left\{ \begin{array}{l} x^2 + y^2 + z^2 = 9 \\ z = 0 \end{array} \right\} = \{ x^2 + y^2 = 9 \} \quad \begin{array}{l} \text{circle} \\ \text{of} \\ \text{radius } 3 \end{array}$$

$$= \{ (x, y, 0) \}$$

$$F|_{C, z=0} = (2y \cdot \cos(z), e^x \sin(z), xe^y)|_{z=0} = (2y, 0, xe^y)$$

$$dr = (dx, dy, 0)$$

Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (2y \cos(z), e^x \sin(z), xe^y)$ and S is the semisphere $x^2 + y^2 + z^2 = 9, z \geq 0$ and oriented upwards.

$$\oint_C F \cdot dr = \int_C 2y \, dx \stackrel{\text{GREEN'S}}{=} \iint_D (0 - 2) \, dA = -2 \iint_D 1 \, dA =$$

$$F \cdot dr = (2y, 0, xe^y) \cdot (dx, dy, 0) = 2y \, dx \quad \boxed{-2 \cdot 9\pi = -18\pi}$$



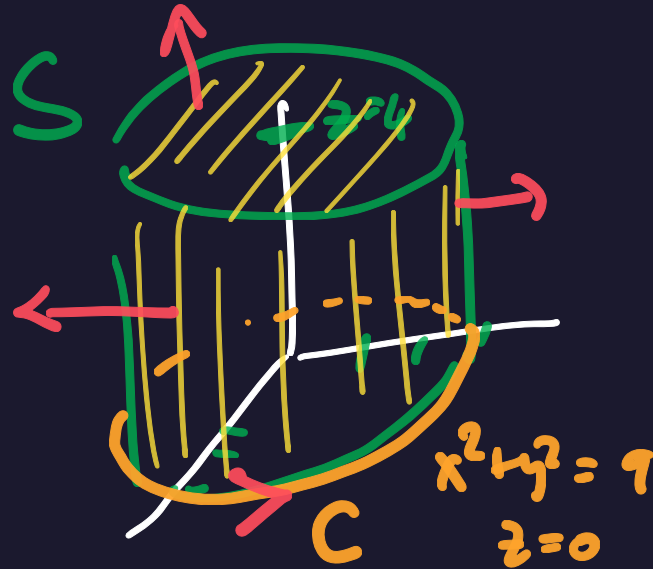
Green's

$$\int P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \, dA$$

$$P = 2y \\ Q = 0$$

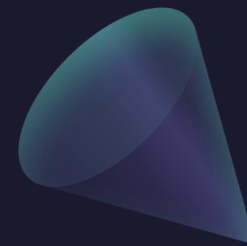
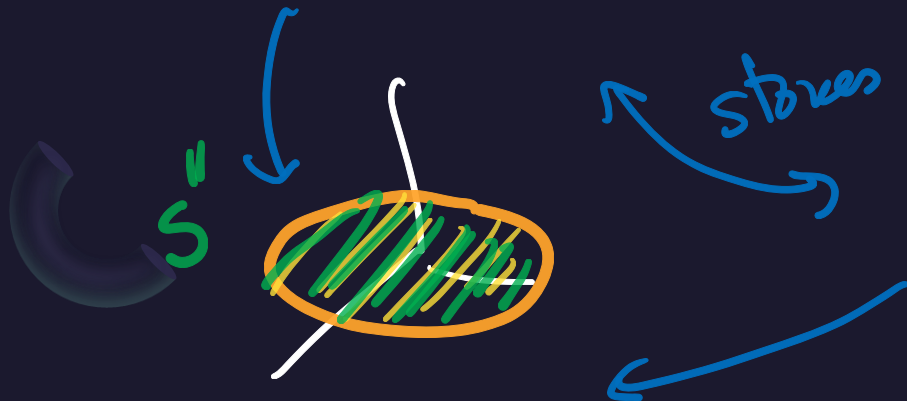
$$\iint_D 1 \, dA = A(D) = 9\pi$$

Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (2y \cos(z), e^x \sin(z), xe^y)$ and S is the cylinder $x^2 + y^2 = 9$, $0 \leq z \leq 4$ along with the top at $z = 4$.

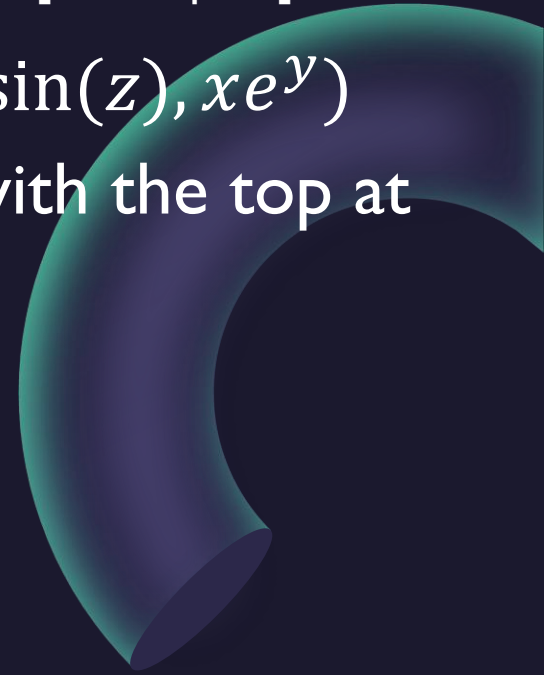
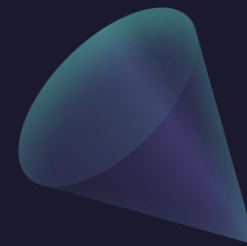


$$\iint_S \text{curl}(F) dS = \iint_{S'} \text{curl}(F) dS$$

$$= \oint_C F dr = \boxed{-18\pi}$$



Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (2y \cos(z), e^x \sin(z), xe^y)$ and S is the cylinder $x^2 + y^2 = 9$, $0 \leq z \leq 4$ along with the top at $z = 4$.



Questions?

$$\vec{F} = (2y, 0, x e^y) \quad C: \begin{matrix} x^2 + y^2 = 9 \\ z = 0 \end{matrix}$$

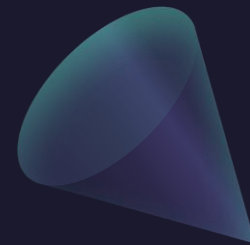
$$\oint_C \vec{F} \cdot d\vec{r}$$

$$\vec{r}(t) = (3 \cos t, 3 \sin t, 0) \quad 0 \leq t \leq 2\pi$$

$$\vec{r}'(t) = (-3 \sin t, 3 \cos t, 0)$$

$$\vec{F}(\vec{r}(t)) = (6 \sin t, 0, 3 \cos t e^{3 \sin t})$$

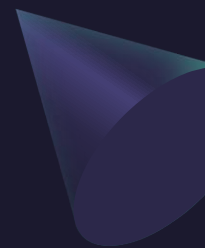
$$\vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) = -18 \sin^2 t$$



$$\oint_C \vec{F} \cdot d\vec{r} = \int_0^{2\pi} -18 \sin^2 t \, dt = -18 \int_0^{2\pi} \sin^2 t \, dt = \boxed{-18\pi}$$



ALVARO: Start the recording!



“Calculus 3”

Multi-Variable Calculus

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The Divergence Theorem

The Divergence Theorem

The Divergence Theorem

$$\nabla \cdot \mathbf{F}$$

$$\operatorname{div} \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

Let E be a simple solid region and let S be the boundary surface of E , given with positive (outward) orientation. Let $\mathbf{F}$ be a vector field whose component functions have continuous partial derivatives on an open region that contains E . Then

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E \operatorname{div} \mathbf{F} \, dV$$

The Divergence Theorem is sometimes called Gauss's Theorem after the great German mathematician Karl Friedrich Gauss (1777–1855), who discovered this theorem during his investigation of electrostatics. In Eastern Europe the Divergence Theorem is known as Ostrogradsky's Theorem after the Russian mathematician Mikhail Ostrogradsky (1801–1862), who published this result in 1826.

Carl Friedrich Gauss



Portrait by [Christian Albrecht Jensen](#), 1840
(copy from Gottlieb Biermann, 1887)^[1]

Born	Johann Carl Friedrich Gauss 30 April 1777 Brunswick , Brunswick- Wolfenbüttel
Died	23 February 1855 (aged 77) Göttingen , Hanover

Example: Find the total flux of the vector field $F = (x^2, 0, 0)$ over the unit sphere $x^2 + y^2 + z^2 = 1$.

$$\iint_S F \cdot ds = \iiint_B \operatorname{div} F \, dv \quad F = (x^2, 0, 0)$$

$$\operatorname{div} F = 2x + 0 + 0 = 2x$$

~~$$- \iiint_{B_2} 2x \, dv$$~~

$$\iiint_B 2x \, dv = \iiint_{B_1} 2x \, dv + \iiint_{B_2} 2x \, dv = 0$$



$$x = a \sin \phi \cos \theta$$

$$y = a \sin \phi \sin \theta$$

$$z = a \cos \phi$$

Example: Find the total flux of the vector field $F = (x^2, 0, 0)$ over the unit sphere $x^2 + y^2 + z^2 = 1$.

$$B = \begin{cases} 0 \leq \rho \leq 1 \\ 0 \leq \theta \leq 2\pi \\ 0 \leq \phi \leq \pi \end{cases}$$

$$\iiint_B 2x \, dV = \iiint = 0$$

spherical ...

$$x = a \sin \phi \cos \theta \quad y = a \sin \phi \sin \theta \quad z = a \cos \phi$$

Example: Find the total flux of the vector field $F = (z, y, x)$ over the unit sphere $x^2 + y^2 + z^2 = 1$.

$$\iint_S F \cdot dS = \iiint_B \operatorname{div} F \, dV$$

DIV THM

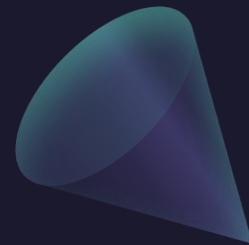
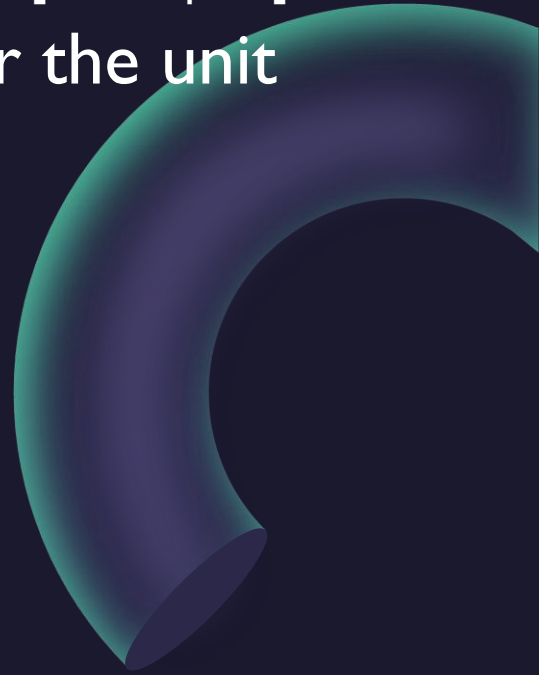
$$F = (z, y, x)$$
$$\operatorname{div} F = 0 + 1 + 0 = 1$$

$$= \iiint_B 1 \, dV = V(B) = \frac{4}{3} \pi (1)^3 = \boxed{\frac{4\pi}{3}}$$

$$\uparrow$$
$$\{x^2 + y^2 + z^2 \leq 1\}$$

$$x = a \sin \phi \cos \theta \quad y = a \sin \phi \sin \theta \quad z = a \cos \phi$$

Example: Find the total flux of the vector field $F = (z, y, x)$ over the unit sphere $x^2 + y^2 + z^2 = 1$.



$$x = a \sin \phi \cos \theta \quad y = a \sin \phi \sin \theta \quad z = a \cos \phi$$

Example: Evaluate $\iint_S F \cdot dS$ where F is the vector field
 $F = (xy, y^2 + e^{xz^2}, \sin(xy))$ and S is the surface of the region
 bounded by $z = 1 - x^2$ and planes $z = 0$, $y = 0$, and $y + z = 2$.



$$\begin{cases} -1 \leq x \leq 1 \\ 0 \leq z \leq 1 - x^2 \\ 0 \leq y \leq 2 - z \end{cases}$$

$$\iint_S F \cdot dS \stackrel{\substack{\uparrow \\ \text{div. thm}}}{=} \iiint_E \operatorname{div} F \, dV$$

div. thm

$$z = 2 - y$$

Example: Evaluate $\iint_S F \cdot dS$ where F is the vector field

$F = (xy, y^2 + e^{xz^2}, \sin(xy))$ and S is the surface of the region bounded by $z = 1 - x^2$ and planes $z = 0$, $y = 0$, and $y + z = 2$.

$$\operatorname{div} F = y + 2y + 0 = 3y$$

$$\oiint_S F \cdot dS = \iiint_E 3y \, dV = \int_{-1}^1 \int_0^{1-x^2} \int_0^{2-z} 3y \, dy \, dz \, dx$$

$$\int_0^{2-z} 3y \, dy = 3 \frac{y^2}{2} \Big|_0^{2-z} = \frac{3}{2} (2-z)^2$$

Example: Find the flux $\iint_S F \cdot d\mathbf{S}$ where F is the vector field $F = (0, y, -z)$ and S is the surface of the region bounded by $y = x^2 + z^2$, $0 \leq y \leq 1$ together with $x^2 + z^2 \leq 1$, $y = 1$.

$$\int_{-1}^1 \int_0^{1-x^2} \frac{3}{2}(2-z)^2 dz dx = \int_{-1}^1 \left[-\frac{1}{2}(2-z)^3 \right]_0^{1-x^2} dx$$

$$\int \frac{3}{2}(2-z)^2 dz = -\frac{1}{2}(2-z)^3$$

$$= \int_{-1}^1 4 - \frac{1}{2}(1+x^2)^3 dx = \dots = \boxed{\frac{184}{35}}$$

Example: Find the flux $\iint_S F \cdot dS$ where F is the vector field $F = (0, y, -z)$ and S is the surface of the region bounded by $y = x^2 + z^2$, $0 \leq y \leq 1$ together with $x^2 + z^2 \leq 1$, $y = 1$.

$$\operatorname{div} F = 0 + 1 - 1 = 0$$

$$\iint_{\cancel{S}} F \cdot dS = 0 = \iiint_E 0 \, dV$$

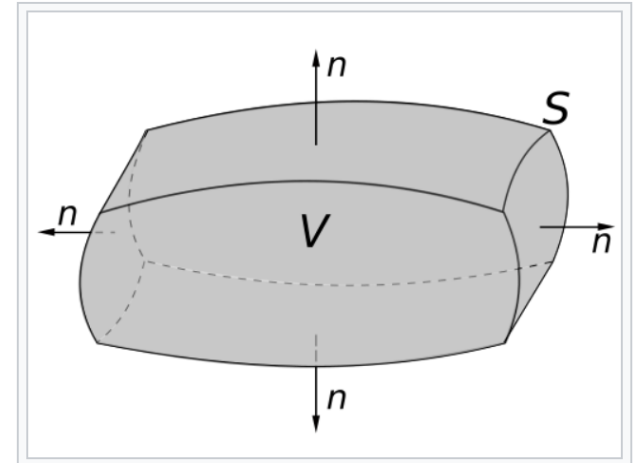
The Divergence Theorem (Gauss's Theorem)

Mathematical statement [\[edit\]](#)

Suppose V is a [subset](#) of $\mathbb{R}^n$ (in the case of $n = 3$, V represents a volume in [three-dimensional space](#)) which is [compact](#) and has a [piecewise smooth boundary](#) S (also indicated with $\partial V = S$). If $\mathbf{F}$ is a continuously differentiable vector field defined on a [neighborhood](#) of V , then:[\[4\]](#)[\[5\]](#)

$$\iiint_V (\nabla \cdot \mathbf{F}) \, dV = \oiint_S (\mathbf{F} \cdot \hat{\mathbf{n}}) \, dS.$$

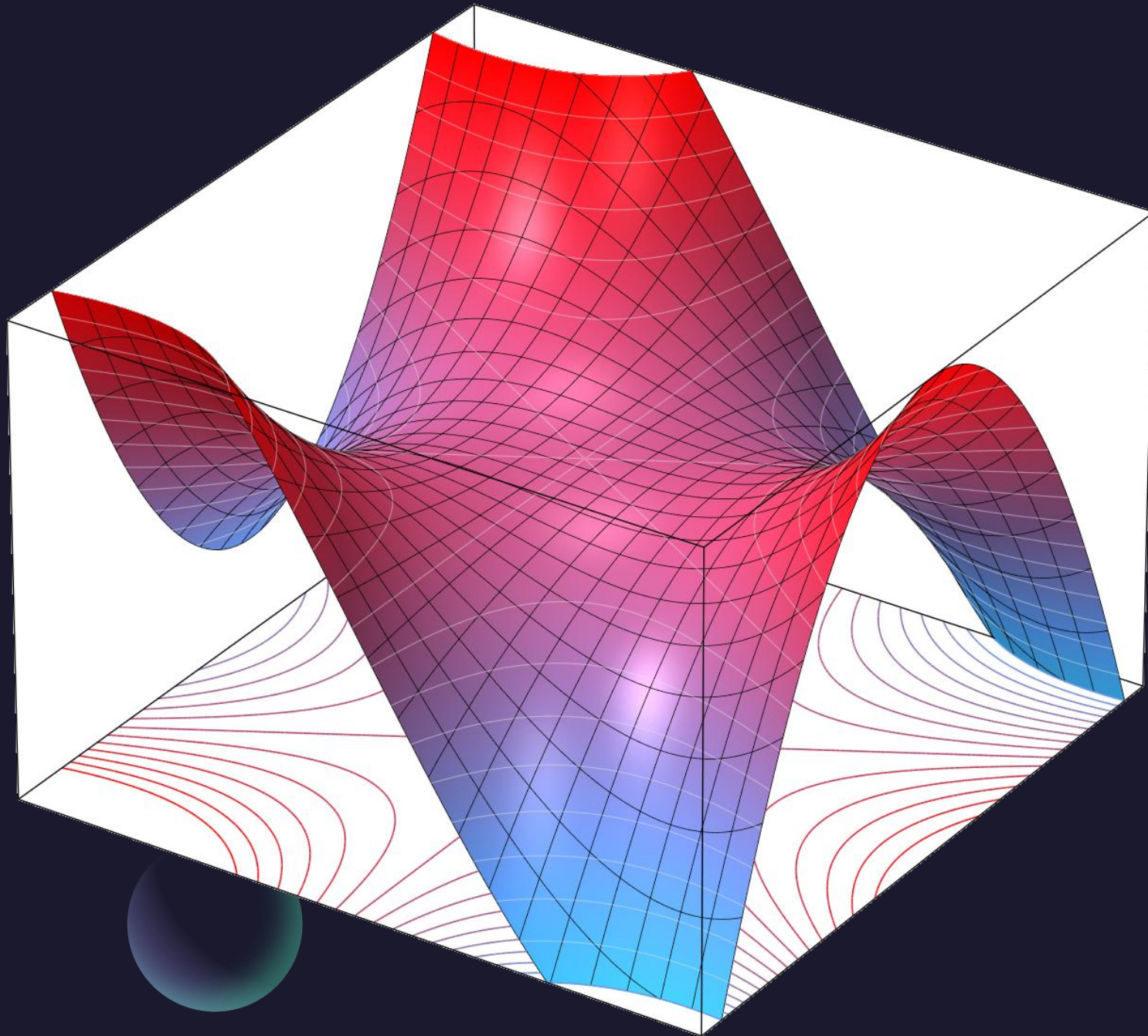
The left side is a [volume integral](#) over the volume V , and the right side is the [surface integral](#) over the boundary of the volume V . The closed, measurable set ∂V is oriented by outward-pointing [normals](#), and $\hat{\mathbf{n}}$ is the outward pointing unit normal at almost each point on the boundary ∂V . ($d\mathbf{S}$ may be used as a shorthand for $\mathbf{n}dS$.) In terms of the intuitive description above, the left-hand side of the equation represents the total of the sources in the volume V , and the right-hand side represents the total flow across the boundary S .



A region V bounded by the surface $S = \partial V$ with the surface normal n

Thank you

Until next time.



Questions?

1. Vector Functions and Curves

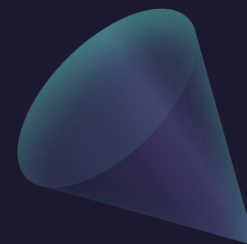
P1. Find the unit tangent vector $\vec{T}(t)$ for $\vec{r}(t) = \langle t, \frac{2}{3}t^{3/2}, t^2 \rangle$ at $t = 1$.

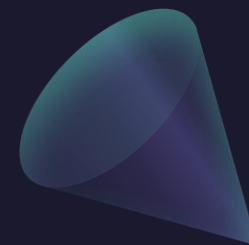
P2. Find the length of the curve $\vec{r}(t) = \langle e^t \cos t, e^t \sin t, e^t \rangle$ for $0 \leq t \leq \ln(2)$.

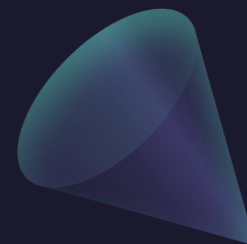
P3. Consider the curve C formed by the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = 1 - y$.

(a) Find a parametrization $\vec{r}(t)$ for C .

(b) Find $\vec{r}'(t)$ and the line integral $\oint_C \langle -y, x, z \rangle \cdot d\vec{r}$.





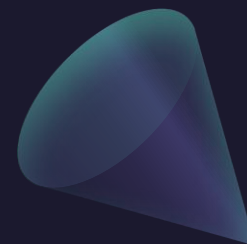


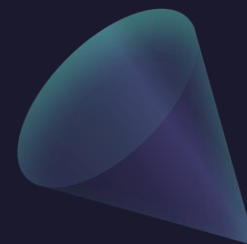
2. Conservative Fields and Green's Theorem

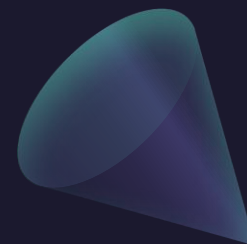
- P4.** Use Green's Theorem to evaluate $\oint_C (xy + e^{x^2}) dx + (x^2 + \sin(y^2)) dy$ where C is the boundary of the rectangle with vertices $(0, 0)$, $(3, 0)$, $(3, 2)$, and $(0, 2)$.
- P5.** Show that $\vec{F} = \langle y \cos(xy), x \cos(xy) + 2y \rangle$ is conservative and find a potential function f . Use it to evaluate $\int_C \vec{F} \cdot d\vec{r}$ from $(0, 0)$ to $(1, \pi)$.

3. Parametric Surfaces and Surface Area

- P6.** Let S be the part of the plane $3x + 2y + z = 6$ in the first octant.
- (a) Find a parametrization $\vec{r}(x, y)$ for S .
 - (b) Express the domain D as a **Type I (Type- x)** domain.
 - (c) Compute $\vec{r}_x \times \vec{r}_y$ and find the surface area of S using integration.
- P7.** Find the area of the surface $z = xy$ that lies within the cylinder $x^2 + y^2 = 4$.







4. Flux and Vector Integrals

P8. Let S be the part of the cone $z = \sqrt{x^2 + y^2}$ between $z = 0$ and $z = 2$.

- (a) Write a parametrization $\vec{r}(r, \theta)$ for S using **polar coordinates**.
- (b) Find the normal vector $\vec{r}_r \times \vec{r}_\theta$.
- (c) Evaluate the flux of $\vec{F} = \langle x, y, z^4 \rangle$ through S with upward orientation.

5. Stokes' and Divergence Theorems

P9. Let $\vec{F} = \langle z^2, x^2, y^2 \rangle$. Use Stokes' Theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$, where C is the triangle with vertices $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$, oriented counterclockwise when viewed from above.

